离散 Lec 08

非 Abel 群 G $\{e\} \triangleleft G_1 \triangleleft G_2 \cdots \triangleleft G$ "没有非平凡正规子群"

G_1 G_2/G_1 G_3/G_2 G_4/G_3 Simple Group: $H \triangleleft G \to H=G$ or $H=\{e\}$

任何有限群都存在分解使得 G_i/G_{i-1} 为单群（若有非 ↓ 则可加入 $\triangleleft$ 里）

Group Action 群作用

$*: G \cdot S \to S$

1) $e * s = s$
2) $(gh) * s = g * (h * s)$

conjuary $G \times G \to G$

$g * x = g x g^{-1}$

↓ G 的子群

$G \times Lattice(G) \to Lattice(G)$

$\{H \mid H \leq G\}$ $g * H = gHg^{-1}$

Orbit $Gx = \{g * x \mid g \in G\} = \dfrac{|G|}{|Stab(x)|}$

Stablizer $Stab(x) = \{g \in G \mid gx = x\}$

$S = \sum |orbit| = \sum \dfrac{|G|}{|Stab(x)|}$ pick an x for each orbit

$= \sum \dfrac{|G|}{|Stab(x)|}$ (orbit size $=1$) $+ \sum \dfrac{|G|}{|Stab(x)|}$ (orbit size >1) $= C_G + \sum_{\cdots x \cdots} [G : C_G(x)]$

$C_G = \{g \mid \forall h \in G \ hg = gh\}$ $C_G(x) = \{g \mid gx = xg\}$

p-Group G means $|G| = p^k$ p means prime

any p-Group G $|C_G| > 1$ Pf. $p \mid |C_G|$

any G $|G| = p^2$ then G Abel

Pf. Assume $|C_G| = p$ pick $g \notin C_G$

$C_G(g) * \supseteq C_G \cup \{g\}$ 矛盾 $\therefore |C_G| = p^2$

Burnside Lemma

$\#\ orbit = \dfrac{1}{|G|} \sum_{g \in G} |S_g|$ $S_g = \{x \in S \mid gx = x\}$

Pf. $= \{(g, x) \in G \times S \mid gx = x\} = \sum_g |S_g|$

$\#$ orbit $|G|$?

$\{(gx) \in G \times S \mid gx = x \wedge x \in G_y\} = |G_y| \cdot |Stab(x_y)| = |G|$

$S_n = Sym(n)$ (123) (1234) (321) $g \cdot (123) g^{-1}$ $(g(1)\ g(2)\ g(3))$

S_4

e	1
(12)	$\binom{4}{2} = 6$
(123)	$2 \cdot \binom{4}{3} = 8$
(1234)	$3! = 6$
$(12)(34)$	$\frac{1}{2} \cdot \binom{4}{2} = 3$
Total	24

p-Subgroup $G = p^k m$

p-Sylow Subgroup $H \leq G$ $|H| = p^k$ 可能的幂次中最大的

Any finite group $\exists$ p-Sylow subgroup

$|G| = p^k m$ Pf. Assume —— for smaller groups

$$|G| = |C_G| + \sum_{c_i} [G : C_G(c_i)]$$

Case $p \nmid |C_G|$ Pick c $p \nmid [G : C_G(c)]$ $p^k \mid C_G(c)$

Case $p \mid |C_G|$ Pick $c \in C_G$ s.t. order(c) = p

$\langle c \rangle \leq C_G$ 因此. 特别地 $\langle c \rangle \trianglelefteq G$

$|G/\langle c \rangle| = p^{k-1} m$ $H/\langle c \rangle \leq G/\langle c \rangle$ s.t. $|H/\langle c \rangle| = p^{k-1}$

Any p-Sylow Subgroup P p-Subgroup Q

$\exists g$ $Q \leq gPg^{-1}$ Pf.

推论 any p-Sylow Subgroup P Q $\exists g$ $Q = gPg^{-1}$

即 any p-Subgroup is contained by a p-Sylow Subgroup

即 p-Sylow Subgroup if Unique iff --- is normal

即 $gP = Pg$ $gPg^{-1} = P$ 即所有 Sylow 子群共轭

Pf. Consider group action

$Q * G/P \to G/P$ $q * gP = qgP$ $\underbrace{|G/P|}_{\text{not divided by p}} = \sum |\text{orbit}|$

$\exists$ orbit (gP) $p \nmid |\text{orbit}(gP)| = [Q : \text{Stab}(gP)] = 1$

$Q = \text{Stab}(gP)$

$= \{q \in Q \mid qgP = gP\}$ $\forall q \in Q.\ qgP = gP$ $g^{-1}qgP = P$

$g^{-1}qg = P$ $q \in gPg^{-1}$ $q \in gPg^{-1}$

即 p-Sylow Subgroup P P is the unique p-Sylow Subgroup of $N_a(P)$

$N_a(P) := \{g \in G \mid gPg^{-1} = P\}$ 正规化子

$P \trianglelefteq N_a(P) \Rightarrow$ 唯一 P-sylow

S_4 $\langle e(12)(34), (13)(24) \rangle$ S_{p^2} $(1 \cdots p)(p+1 \cdots 2p) \cdots (p^2-1\ p+1 \cdots p^2)$

$Sylow_p(G) = \{\text{all } p\text{-Sylow subgroup}\}$

$|G| = p^r m$ Let $n_p = |Sylow(G)|$ Then $n_p | m$ & $n_p \equiv 1 \bmod p$

Pf. of $n_p | m$.

$G \times Subgroup(G) \to Sylow(G) \quad g \times P \mapsto gPg^{-1}$

$n_p = |Sylow(G)| = [G : N_G(P)] = \frac{|G|}{|N_G(P)|} = \frac{|G|}{|P| \cdot [N_G(P) : P]} = \frac{m}{[N_G(P) : P]}$

$\{g \in G \mid gPg^{-1} = P\}$ $\qquad p^r$

Pf. $n_p \equiv 1 \bmod p$

Pick p-Sylow Subgroup P $\qquad P \times Sylow_p(G) \to Sylow_p(G)$

$g \times P_i \mapsto gP_ig^{-1}$ $\qquad n_p = \Sigma |\text{orbit}| = \Sigma [P_i : Stab(P_i)]$

① $Stab(P_i) \neq P$ $\quad p \mid |orbit|$

② $Stab(P_i) = P \to P \subseteq N_G(P_i)$ $\quad P = P_i$ $|orbit| = 1$

Group G $|G| = 35$ then G Abel

$n_5 | 7$ $\quad n_5 \equiv 1 \bmod 5$

$n_7 | 5$ $\quad n_7 \equiv 1 \bmod 1$ $\Rightarrow n_5 = n_7 = 1$

$\mathbb{Z}_5 \cong C_5 \trianglelefteq G$

$\mathbb{Z}_7 \cong G \trianglelefteq G$

$C_5 C_7 \cdot C_5 \times C_7$

$\| \qquad \|$

$C_1 \qquad \mathbb{Z}_{35}$

$|G| = 11^2 \times 13^2$

then G Abel

$n_{11} | 13^2$ $\quad n_{11} \equiv 1 \bmod 11$ $\quad n_{11} = 1$

$n_{13} | \quad n_{13} = 1$

$C_{11} \trianglelefteq G \quad C_{11}$ Abel

$C_{13} \trianglelefteq G \quad C_{13}$ Abel

Group G $|G| = pq$ $q > p$

$n_q | p$ $\quad n_q \equiv 1 \bmod q$ $\Rightarrow n_q = 1$ $\quad q$-Sylow Subgroup $\langle g \rangle \trianglelefteq G$

$n_p | q$ $\quad n_p \equiv 1 \bmod p$ $\quad n_p = 1$ or q $\quad p$-Sylow Subgroup $\langle h \rangle \leq G$

① $q \not\equiv 1 \bmod p$ G Abel ② $q \equiv 1 \bmod p$

$hgh^{-1} \in \langle g \rangle = g^s$ $\quad s \neq 0$ ~~if $s = 1$~~ $G = \langle h \times g \rangle = \{h^x g^y\}$

if $s = 1$ G Abel

if $s \neq 0, 1$ $\quad \langle h^x g^y \mid g^q = h^p = 1 \quad hgh^{-1} = g^s \rangle$

$\langle g \rangle = \langle g^t \rangle$ $\quad q \nmid t$ $\quad \hat{g}^n = g^t$ $\quad h\hat{g}h^{-1} = (hgh^{-1})^t =$

$\hat{h} = ht$ $\quad \hat{h} g \hat{h}^{-1} = g^{st}$ $\quad \langle \hat{h}^x g^y \mid g^q = \hat{h}^p = 1 \rangle$

$\hat{h} g \hat{h}^{-1} = g^{st}$

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